



AtliQ Motors

Resume Project Challenge 12



EV MARKET ANALYSIS IN INDIA



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Problem Statement

Background

AtliQ Motors is a renowned automotive company in the USA with a stronghold in the electric vehicle (EV) and hybrid market. Over the past 5 years, their market share has increased to 25% in North America.

Challenge

As part of its global expansion strategy, AtliQ Motors plans to enter the Indian market, where its current market share is less than 2%.

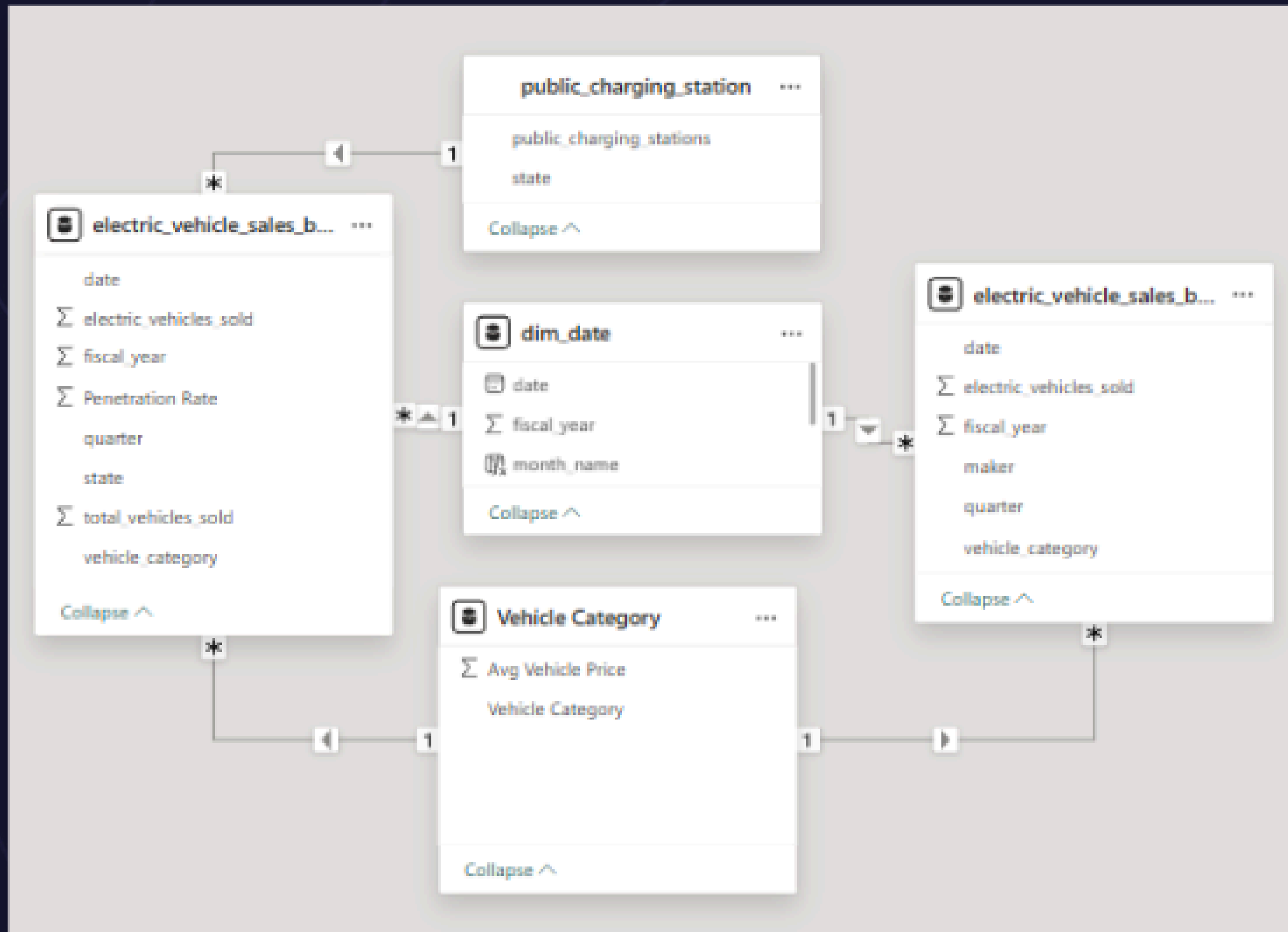
Before launching their bestselling EV models in India, AtliQ Motors needs to conduct a thorough market analysis to understand the existing EV/Hybrid landscape in India.



Goal

Objective

The primary goal is to provide AtliQ Motors with a comprehensive market analysis of the Indian EV and hybrid vehicle market. This analysis will guide their market entry strategy, helping the company to make informed decisions about where and how to launch their vehicles in India.



Data Familiarization

- Tables of the input data.
- Input data consists of sales data for FY 2022, 2023 & 2024

Tool Used

PowerBI



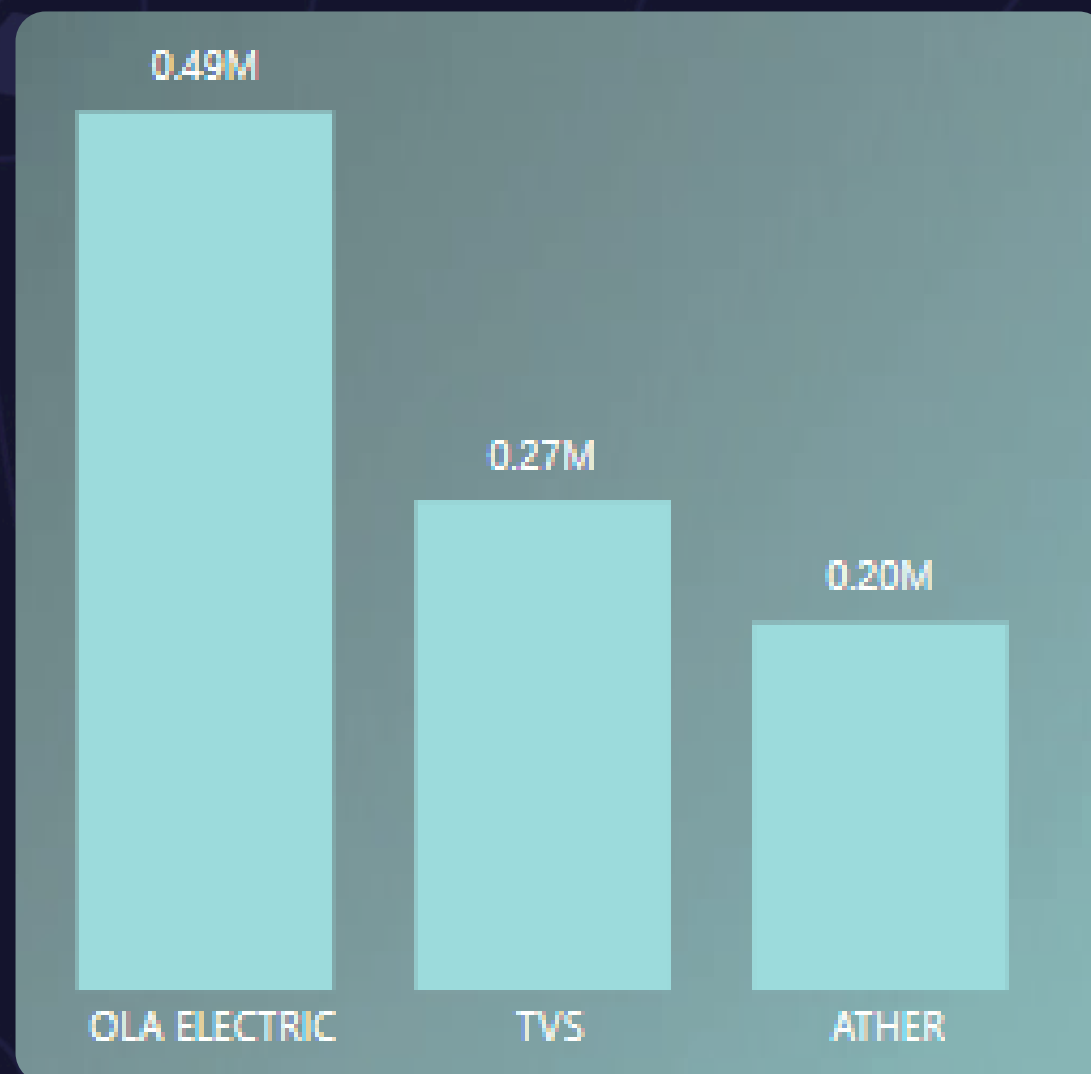


Primary Research Questions

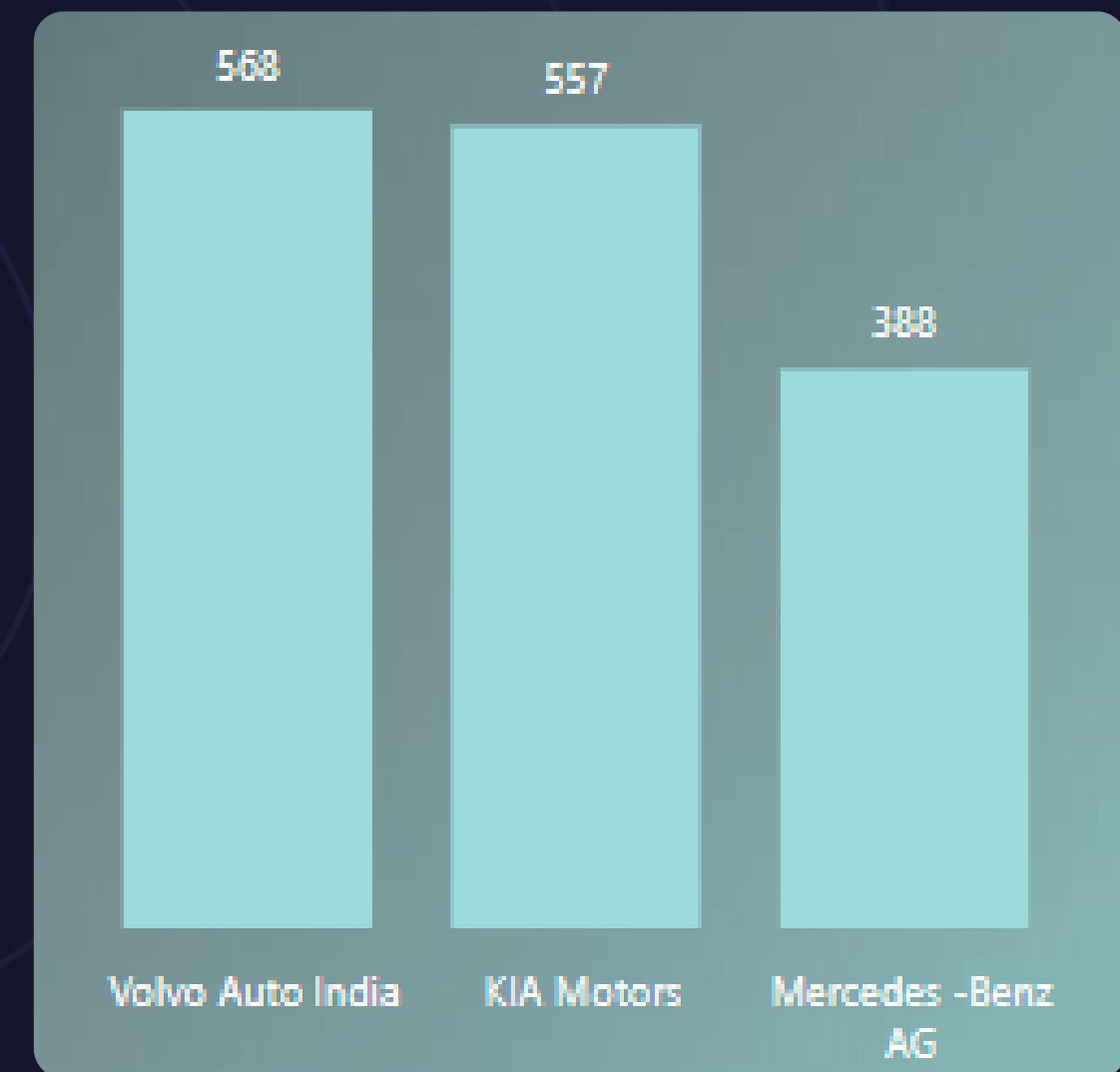


Q1.

List the top 3 and bottom 3 makers for the fiscal years 2023 and 2024 in terms of the number of 2-wheelers sold.

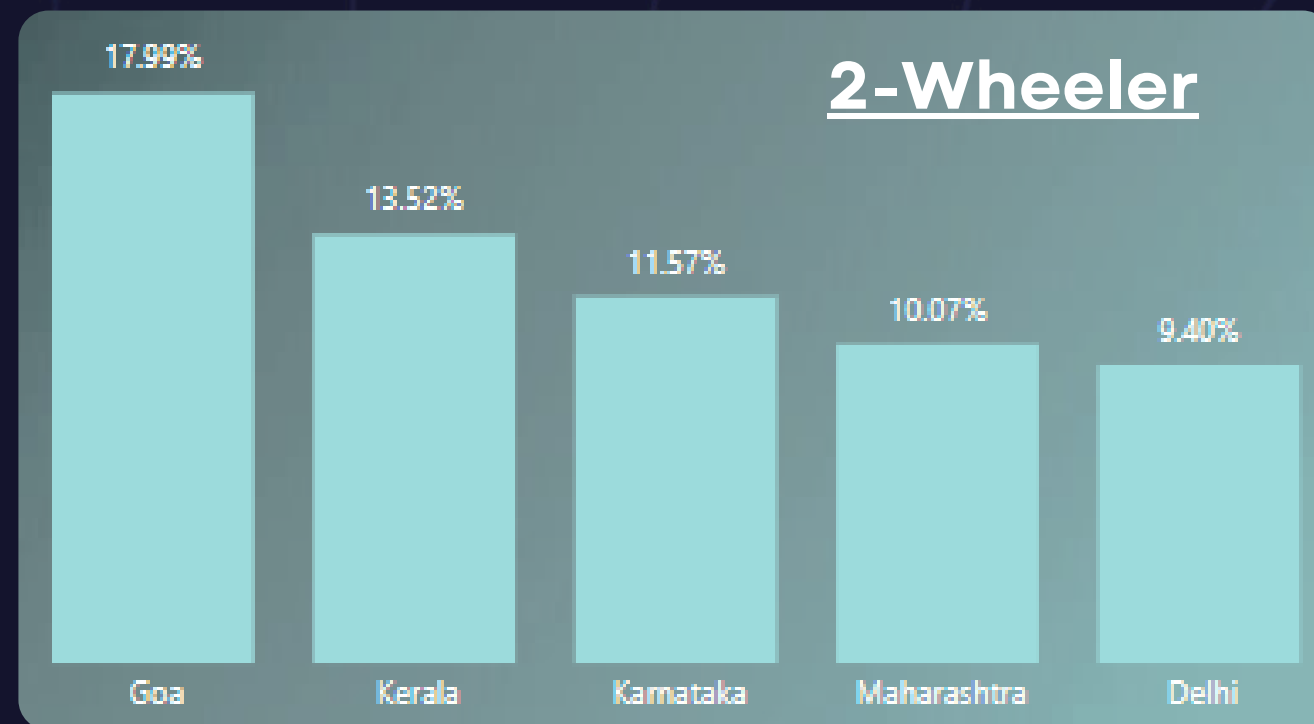


Top 3 Makers

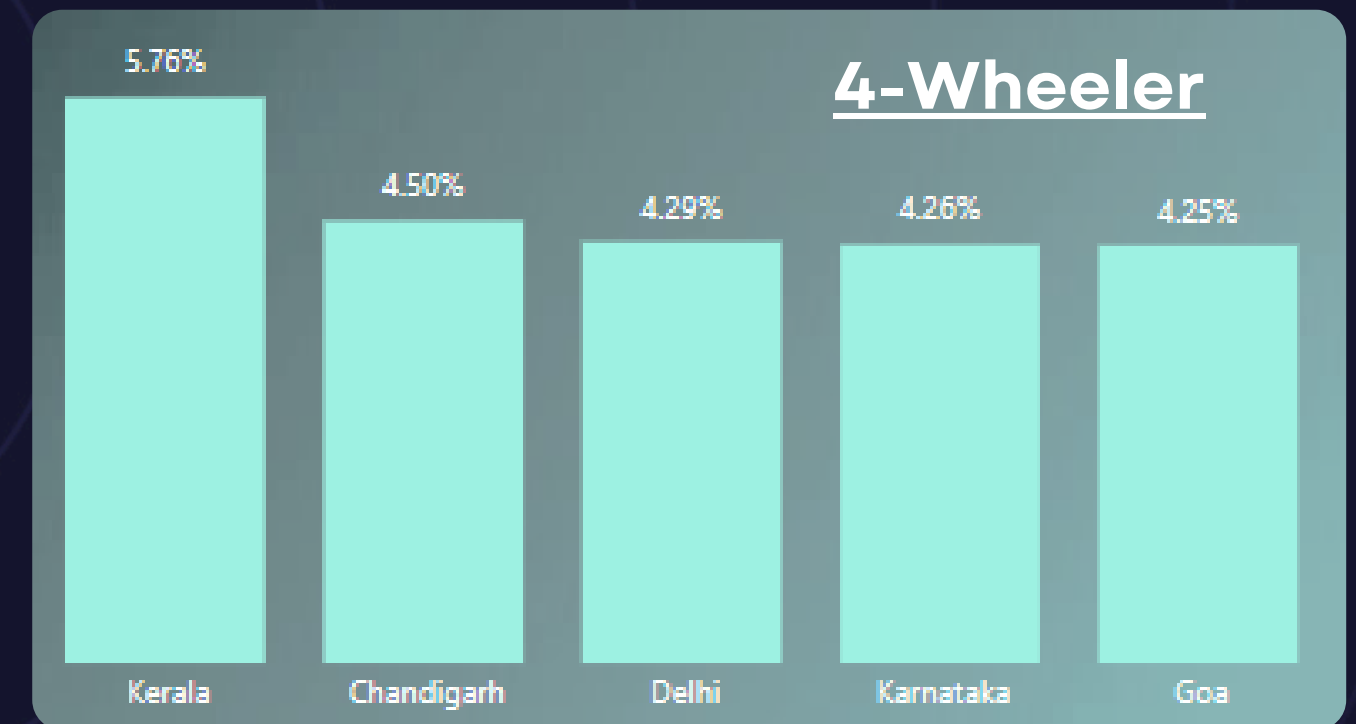


Bottom 3 Makers

Q2.
Identify the top 5 states with the highest penetration rate in 2-wheeler and 4-wheeler EV sales in FY 2024.



Top 5 States



Bottom 5 States

Q3.
List the states with negative penetration (decline) in EV sales from 2022 to 2024?

STATE	PR% 2022	PR% 2024	PR% 2022vs24
Sikkim	0.00%	0.00%	0.00%
Nagaland	0.01%	0.05%	0.05%
Andaman & Nicobar Island	0.43%	0.49%	0.06%
Arunachal Pradesh	0.00%	0.11%	0.11%
Meghalaya	0.02%	0.36%	0.35%
Himachal Pradesh	0.45%	0.90%	0.44%
Assam	0.19%	0.64%	0.45%
Haryana	1.12%	1.61%	0.49%
Ladakh	0.41%	0.97%	0.55%

No decline in PR%
(No EV Vehicle Category
Selected)

2-Wheeler

STATE	PR% 2022	PR% 2024	PR% 2022vs24
Ladakh	4.48%	4.06%	-0.41%

Decline of -0.41% in PR% for EV 2W

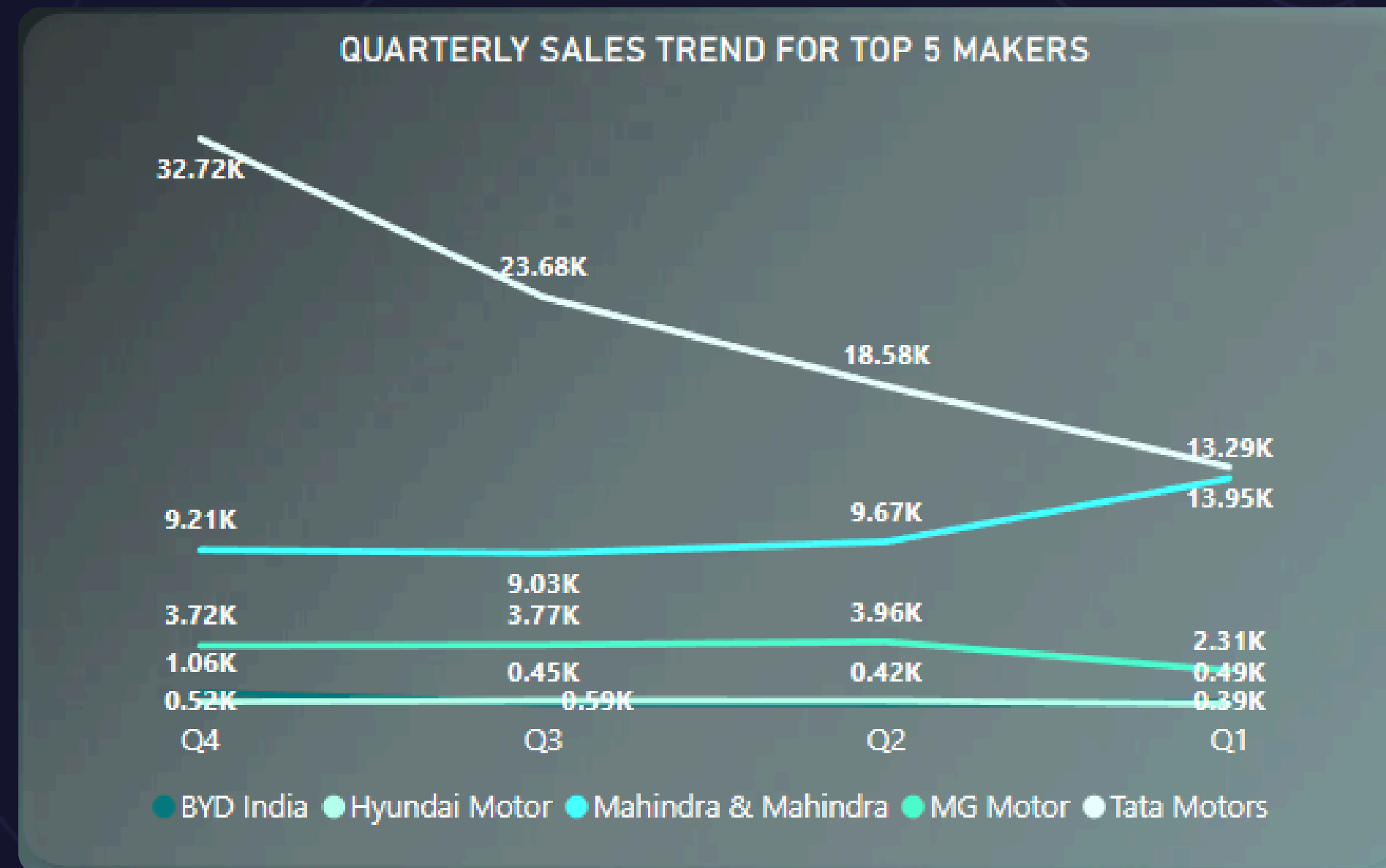
4-Wheeler

STATE	PR% 2022	PR% 2024	PR% 2022vs24
Andaman & Nicobar Island	1.88%	0.77%	-1.11%

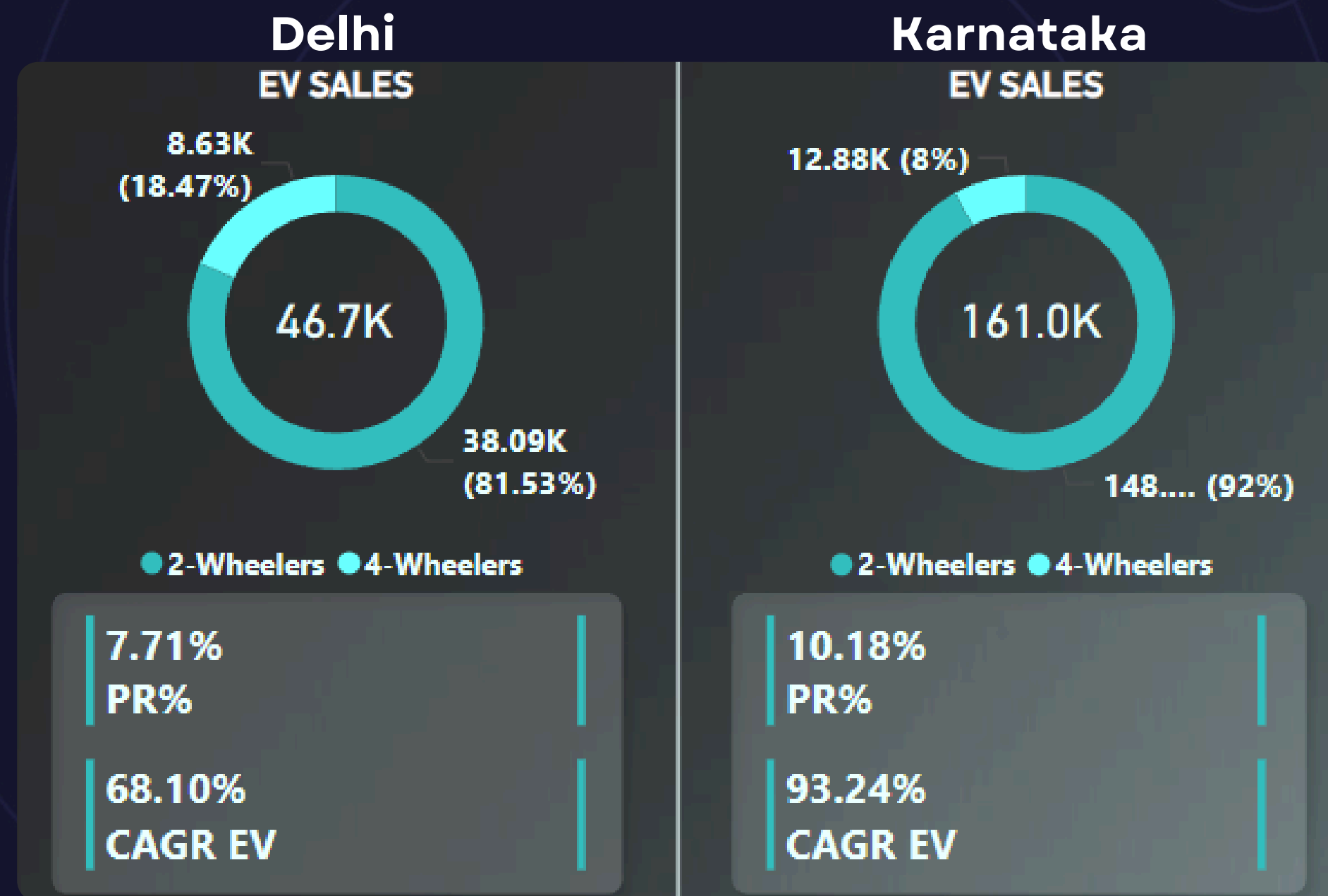
Decline of -1.11% in PR% for EV 4W

Q4.

What are the quarterly trends based on sales volume for the top 5 EV makers (4-wheelers) from 2022 to 2024?



Q5.
How do the EV sales and penetration rates in Delhi compare to Karnataka for 2024?



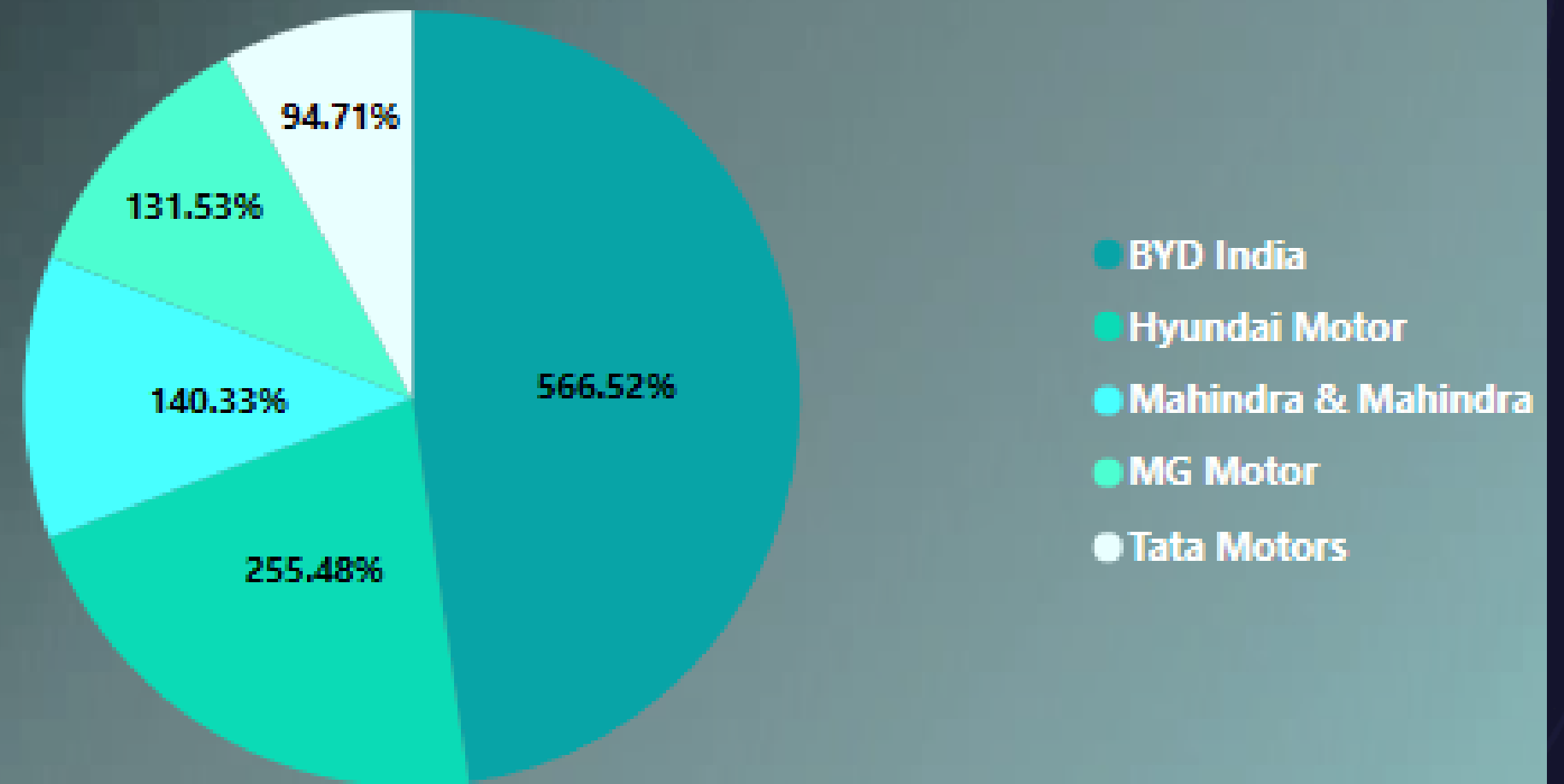
Q6.

List down the compounded annual growth rate (CAGR) in 4-wheeler units for the top 5 makers from 2022 to 2024.

Compound Annual Growth Rate (CAGR): CAGR measures the mean annual growth rate over a specified period longer than one year.

$$\text{CAGR} = \left(\frac{\text{Final value}}{\text{Starting value}} \right)^{\frac{1}{N}} - 1$$

MAKER	CAGR EV
BYD India	566.52%
Hyundai Motor	255.48%
Mahindra & Mahindra	140.33%
MG Motor	131.53%
Tata Motors	94.71%



Q7.

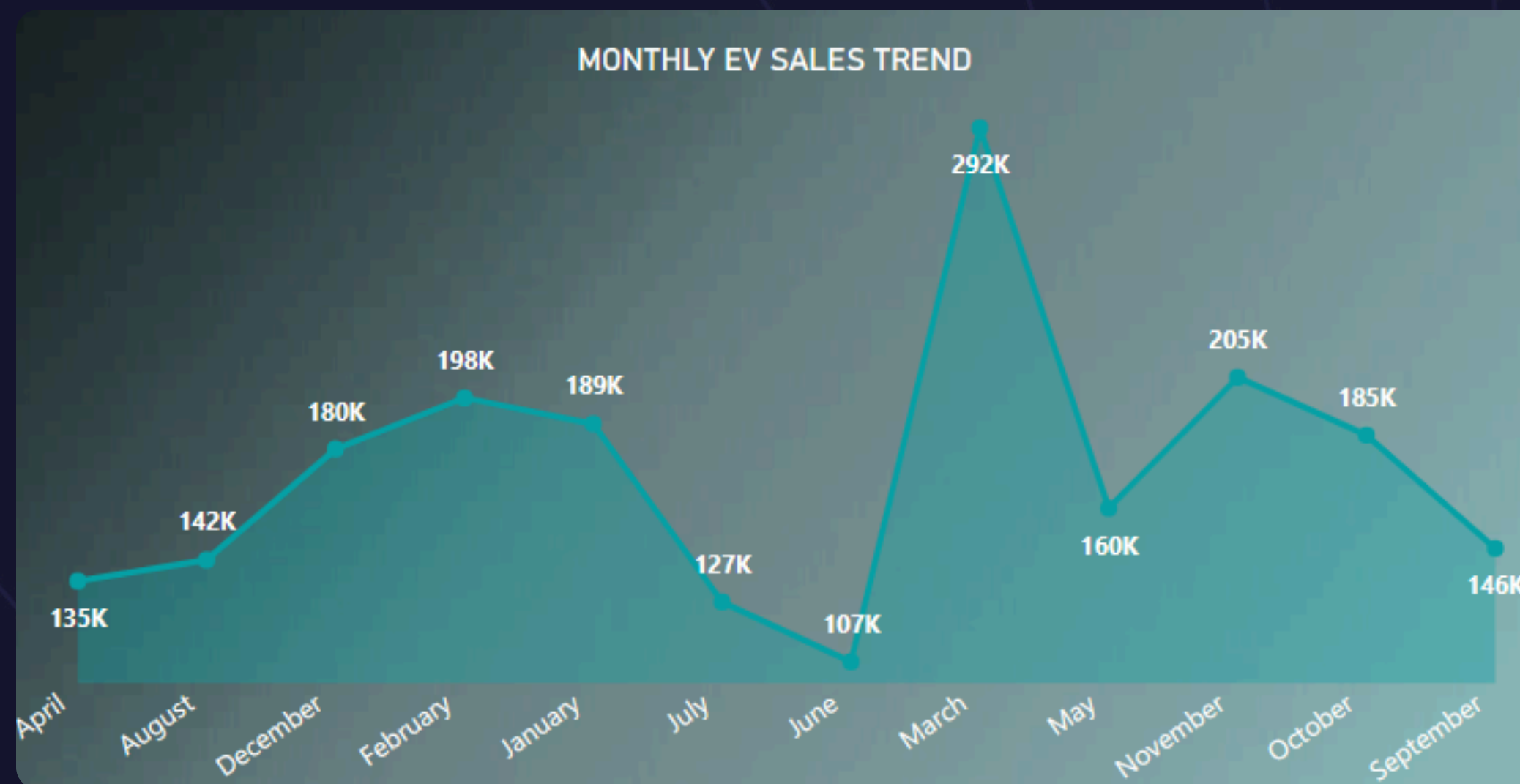
List down the top 10 states that had the highest compounded annual growth rate (CAGR) from 2022 to 2024 in total vehicles sold.

STATE	CAGR TV
Meghalaya	28.47%
Goa	27.41%
Karnataka	25.28%
Delhi	22.88%
Rajasthan	21.50%
Gujarat	20.55%
Assam	20.13%
Mizoram	18.77%
Arunachal Pradesh	18.30%
Andaman & Nicobar Island	18.29%

Q8.

What are the peak and low season months for EV sales based on the data from 2022 to 2024?

The peak season month for EV sale is March: 292K
&
The low season month for EV sale is June : 107K

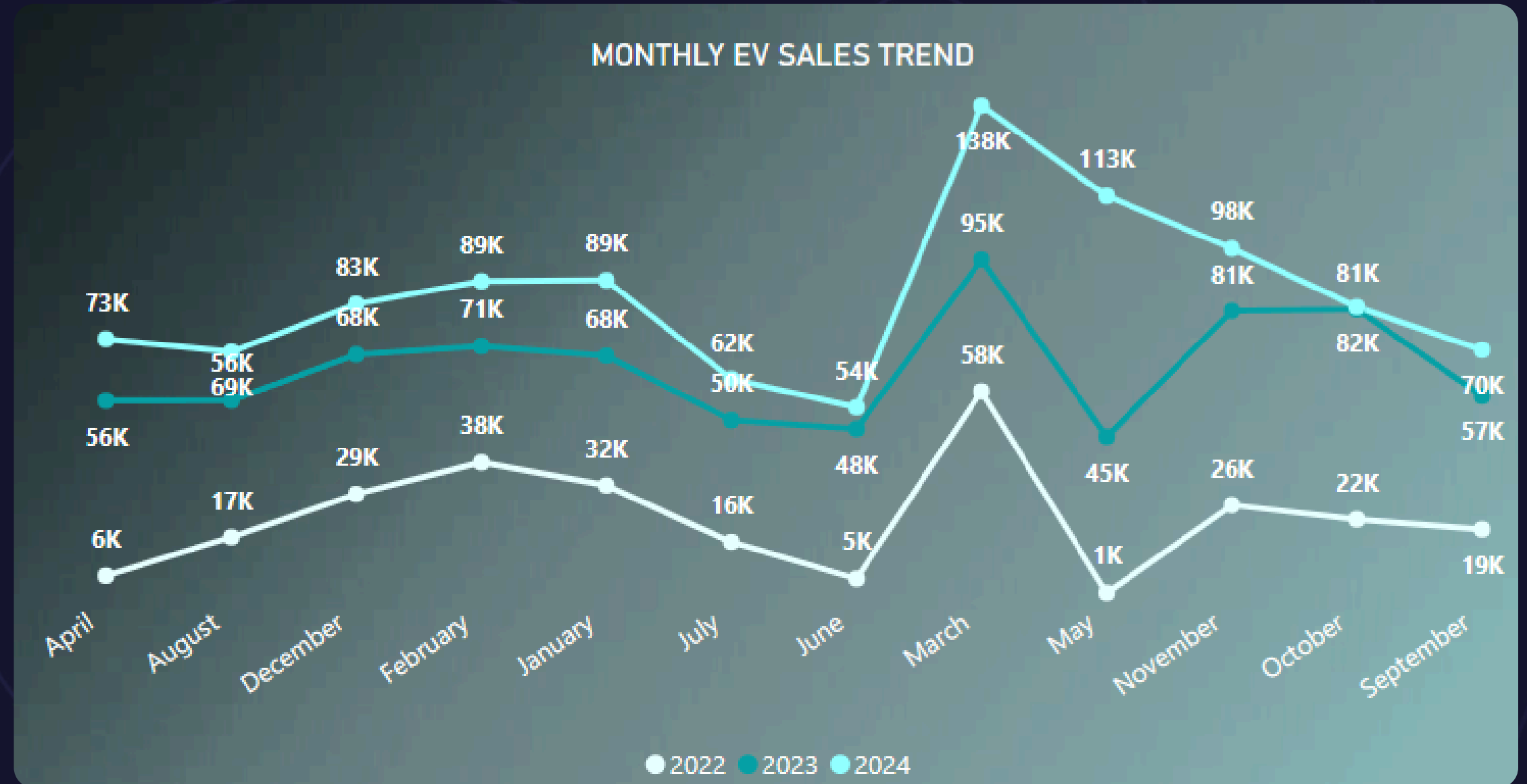




The peak season month for EV sale for 2022, 2023 & 2024 is March

&

The low season month for EV sale for 2022 & 2023 is May,
for 2024 is June



Q9.

What is the projected number of EV sales (including 2-wheelers and 4-wheelers) for the top 10 states by penetration rate in 2030, based on the compounded annual growth rate (CAGR) from previous years?

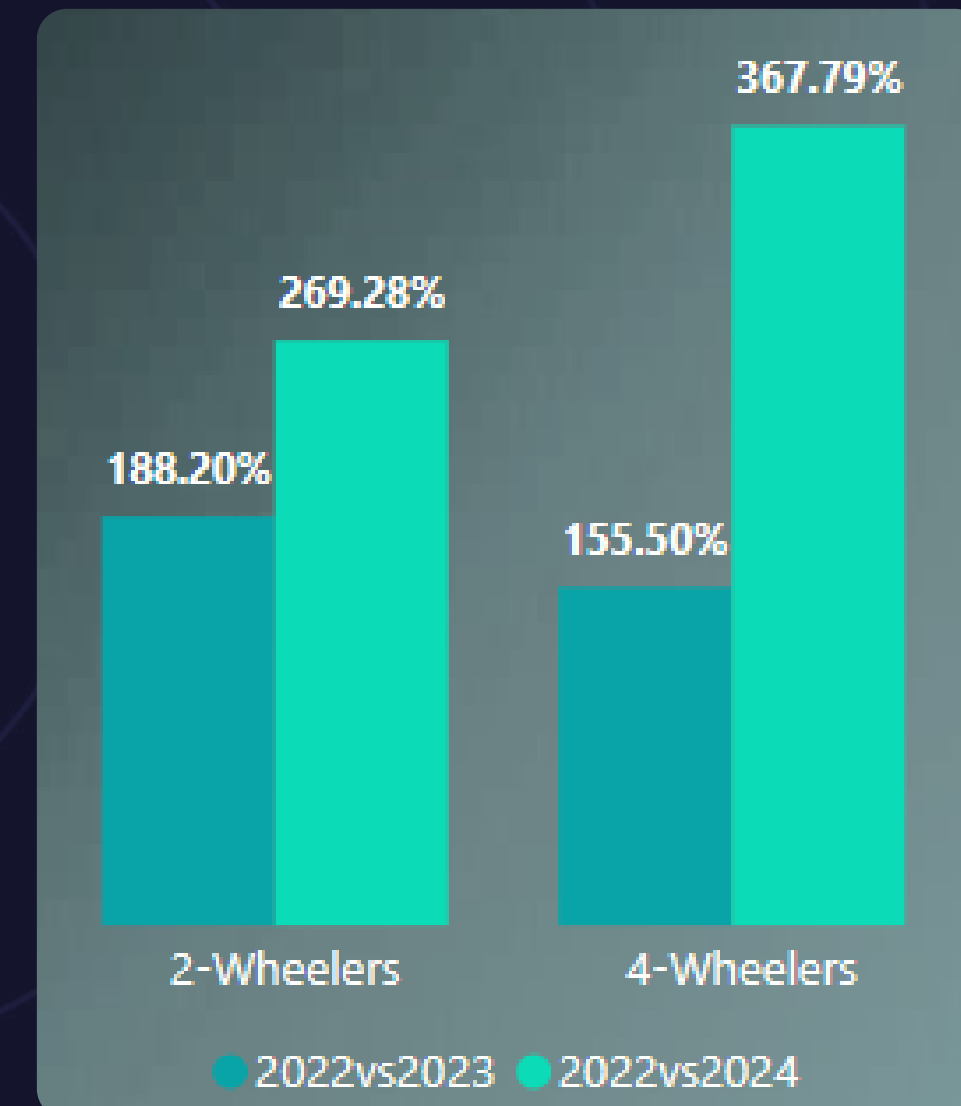
Maharashtra & Kerala have the highest Projected Sales



Q10.

Estimate the revenue growth rate of 4-wheeler and 2-wheelers EVs in India for 2022 vs 2024 and 2023 vs 2024, assuming an average unit price.

VEHICLE CATEGORY	AVG UNIT PRICE
2-Wheelers	85000
4-Wheelers	1500000





Secondary Research Questions





Q1.

What are the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024 (cost savings, environmental concerns, government incentives)?

- **Cost Savings:** EVs have lower operating costs due to cheaper electricity compared to fuel and reduced maintenance needs. Government incentives, such as the FAME II subsidy and state-specific benefits, significantly lower the initial purchase cost.
- **Environmental Concerns:** With growing awareness of climate change, consumers are more conscious of their carbon footprint. EVs produce zero tailpipe emissions, helping to reduce air pollution, especially in urban areas.
- **Government Incentives:** Tax credits and rebates lower the upfront costs of EVs, making them more financially accessible.
- **Technological Advancements:** Improved range (200-300 km per charge) and expanding charging infrastructure reduce range anxiety. Innovations in battery technology and vehicle performance make EVs more attractive.
- **State-Specific Benefits:** Additional financial incentives, such as registration fee waivers and road tax exemptions in states like Maharashtra, Delhi, and Gujarat, make EVs even more affordable.

Q2.

How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?

Government Support for EV Adoption

Key Schemes:

- FAME India Scheme: Direct subsidies for electric 2-wheelers and 4-wheelers, making them more affordable.
- GST Reductions: GST on electric vehicles cut from 12% to 5%, and on EV charging stations from 18% to 5%, reducing ownership costs.
- Extended Subsidies: Financial support for electric two-wheelers and three-wheelers extended until September 2023.

Top Subsidy States:

- Maharashtra: Offers the highest subsidies, leading in EV adoption.
- Other Key States: Delhi, Gujarat, Karnataka, and Tamil Nadu also provide significant subsidies, boosting accessibility and adoption.

Q3.

How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?

STATE	EV SALES	PCS
Maharashtra	396045	3079
Karnataka	312995	1041
Tamil Nadu	200062	643
Gujarat	181389	476
Rajasthan	150366	500

- **More Charging Stations = More EVs Sold:** States with better charging infrastructure, like Maharashtra and Karnataka, sell more EVs.

STATE	PR%	PCS
Goa	9.84%	113
Karnataka	7.84%	1041
Delhi	6.76%	1886
Kerala	6.64%	852
Maharashtra	6.49%	3079

- **Higher EV Penetration:** States like Karnataka and Delhi with more charging stations have higher rates of EV adoption.

STATE	EV SALES	Penetration Rate	PCS
Maharashtra	396045	6.49%	3079
Delhi	107312	6.76%	1886
Karnataka	312995	7.84%	1041
Kerala	137060	6.64%	852
Tamil Nadu	200062	4.30%	643

In summary, there is a strong positive correlation between the availability of charging stations and the EV sales and penetration rates in the top 5 states.

Key Point: States that invest in charging stations see more EV sales and higher adoption rates.

Q4.

Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?



Virat Kohli

Popularity & Credibility:

- As one of India's most popular and influential sports personalities with a massive fan following, which can help AtliQ Motors gain recognition. As a highly respected athlete, his endorsement would boost brand recognition and credibility.

Positive Image & Youth Appeal:

- Virat Kohli has a clean and positive public image which make him a highly appealing figure to young people, which can positively impact AtliQ Motors' brand image and attract a younger customer base.

Alignment with brand values:

Virat Kohli passion for fitness and health aligns perfectly with the eco-friendly nature of electric vehicles. His association with the brand would resonates with the consumers who are conscious about environment.

Q5.

Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)



Maharashtra

Subsidies: Up to ₹25,000 for electric two-wheelers, ₹30,000 for three-wheelers, and ₹2.5 lakh for four-wheelers. Zero registration charges and road taxes for electric three-wheelers.

Ease of doing Business & Market Access: Maharashtra ranks high in ease of doing business indices. Large consumer market for EVs and strategic location for regional and international trade. Access to a large pool of skilled labor in the automotive industry.

Infrastructure & Skilled Workforce: Well-developed infrastructure, including roads, railways, and ports.

Government Support: Supportive government policies and initiatives for the EV industry.





Q6.

Your top 3 recommendations for AtliQ Motors.

Prioritize two-wheeler EVs and utilize government incentives:

Capitalize on the growing demand for affordable electric two-wheelers in India and leverage subsidies and tax benefits to enhance market competitiveness.

Invest in cutting-edge battery technology:

Focus on research and development to improve battery efficiency, lifetime, and cost-effectiveness, ensuring AtliQ Motors stays ahead of the competition.

Expand charging infrastructure and offer comprehensive services: Partner with local entities to increase the availability of charging stations and provide exceptional after-sales support to enhance customer satisfaction and loyalty.



Thank you

